PROJECT REPORT

1.INTRODUCTION

1.1 Project Overview

Project: Electric Vehicle Analysis using Tableau

The rapid adoption of Electric Vehicles (EVs) has transformed the automobile industry by promoting sustainable transportation and reducing environmental pollution. With the increasing availability of EV models, advancements in battery technology, and the expansion of charging infrastructure, large volumes of EV-related data are being generated. However, extracting meaningful insights from this data can be challenging due to its complexity and the wide range of attributes such as vehicle range, price, efficiency, charging stations, body style, and powertrain type. There is a growing need for an interactive analytical platform that converts raw EV data into clear and meaningful visual insights for better understanding and decision-making.

This project presents **Electric Vehicle Analysis using Tableau**, an interactive business intelligence solution designed to analyze and visualize Electric Vehicle data through dynamic dashboards and storytelling. The project explores key aspects of the EV market, including vehicle pricing, driving range, energy efficiency, charging infrastructure, body styles, powertrain types, and regional distribution of charging stations across India. By utilizing Tableau's interactive dashboards, KPI cards, filters, calculated fields, and data storytelling features, users can efficiently explore market trends, compare EV models, and gain valuable insights through intuitive visualizations.

The project includes an interactive dashboard containing **eight analytical visualizations**, **four KPI cards**, and a **Tableau Story** that present the findings in a structured and user-friendly manner. The dashboard provides insights into EV models, average range, average price, average efficiency, charging station distribution, vehicle performance, and brand comparisons. The completed dashboard and story are published on **Tableau Public**, allowing users to access and interact with the visualizations online from anywhere.

The insights generated through this project can assist **customers, researchers, automobile manufacturers, policymakers, and students** in understanding the current Electric Vehicle ecosystem and identifying market trends. By transforming raw EV data into meaningful visual analytics, the project supports data-driven decision-making, promotes awareness of sustainable transportation, and demonstrates the practical application of business intelligence and data visualization techniques in the electric mobility sector.

Scenario 1: Customer Selecting an Electric Vehicle
A customer wants to purchase an electric vehicle but is unsure which model best suits their budget and driving requirements. Using the dashboard, the customer compares EV models based on price, driving range, top speed, energy efficiency, and body style. The insights help them choose the most suitable electric vehicle.

Scenario 2: Government/Charging Infrastructure Planner

A government official analyzes the distribution of charging stations across different regions of India using the dashboard. By identifying areas with fewer charging stations and understanding the available charger types, they can make informed decisions about where to expand the EV charging infrastructure.

1.2 Purpose

The primary purpose of this project is to develop an interactive data visualization platform for analyzing Electric Vehicle (EV) data using Tableau. The project aims to transform raw EV datasets into meaningful visual insights that help users understand vehicle performance, pricing, driving range, energy efficiency, charging infrastructure, and market trends through interactive dashboards and storytelling.

The project is designed to simplify the exploration of complex EV data by presenting information through charts, maps, KPI cards, filters, and Tableau Stories. It enables users to compare different electric vehicle models, identify the distribution of charging stations across India, evaluate vehicle efficiency and pricing, and analyze brand performance based on various parameters.

2.IDEATION PHASE

The ideation phase focused on identifying the challenges involved in analyzing Electric Vehicle (EV) data and exploring suitable analytical solutions. Various user requirements and stakeholder perspectives were considered to develop an interactive and user-friendly dashboard. Brainstorming, empathy mapping, and problem analysis techniques were used to refine the project idea. Based on these insights, an **Electric Vehicle Analysis using Tableau** solution was selected. This phase established the foundation for developing interactive dashboards, KPI cards, and Tableau Stories that support data-driven decision-making.

2.1 Problem Statement

The increasing adoption of Electric Vehicles (EVs) has resulted in the generation of large volumes of data related to vehicle performance, pricing, charging infrastructure, and efficiency. However, this information is often scattered across multiple sources, making it difficult to analyze and compare effectively. Traditional analysis methods are time-consuming and do not provide clear visual insights for decision-making.

This project addresses these challenges by developing an **Electric Vehicle Analysis Dashboard using Tableau**. The solution transforms raw EV data into interactive dashboards, KPI cards, maps, charts, and Tableau Stories, enabling users to analyze EV trends, compare vehicle models, evaluate charging infrastructure, and make informed, data-driven decisions.

Fig 2.1: Problem Statement Canvas

Ideation Phase
Define the Problem Statements

Date	12 July 2026
Team ID	
Project Name	Electric Vehicle Charge and Range Analysis
Maximum Marks	2 Marks

Customer Problem Statement Template:

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.

A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

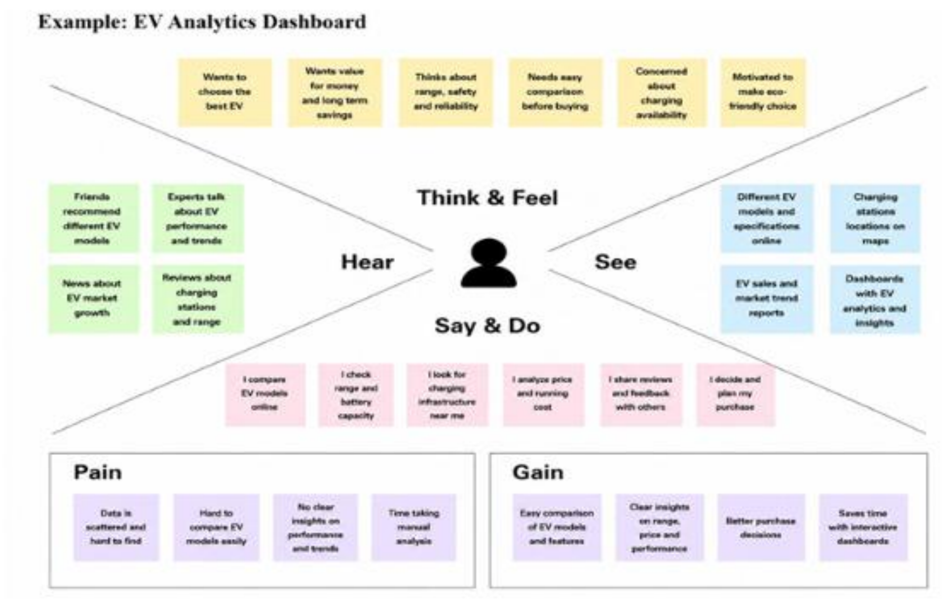
I am	Describe customer with 3-4 key characteristics - who are they?	Describe the customer and their attributes here
I'm trying to	List their outcome or "job" the care about - what are they trying to achieve?	List the thing they are trying to achieve here
but	Describe what problems or barriers stand in the way – what bothers them most?	Describe the problems or barriers that get in the way here
because	Enter the "root cause" of why the problem or barrier exists – what needs to be solved?	Describe the reason the problems or barriers exist
which makes me feel	Describe the emotions from the customer's point of view – how does it impact them emotionally?	Describe the emotions the result from experiencing the problems or barriers

Reference: https://miro.com/templates/customer-problem-statement/ **Example:**

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	EV Buyer	Find the best EV Model	Hard to compare EVs	Data is scattered	Confused
PS-2	EV Analyst	Analyze EV trends	Insights are difficult to find	No Interactive visualization	Time-Consuming

2.2 Empathy Map

The Empathy Map illustrates the perspectives of the primary stakeholders, including **EV buyers, researchers, and policymakers**. It captures what they **think, observe, say, and do** while interacting with EV data, along with their challenges and expectations. These insights guided the development of interactive dashboards, KPI cards, filters, and Tableau Story to ensure the solution effectively supports EV market analysis, charging infrastructure evaluation, and data-driven decision-making.



2.2 Empathy Map Canvas

Based on the sample report, your **Empathy Map Canvas** should have **three stakeholders**, each with **Says, Thinks, Does, and Feels**. Here is the content for your **Electric Vehicle Charge & Range Analysis** project.

1. Rahul – EV Buyer

Says

- "Which EV offers the best range within my budget?"
- "Where can I find nearby charging stations?"

Thinks

- Wants an affordable EV with good range and high efficiency.
- Needs reliable information before purchasing.

Does

- Compares EV models, prices, range, and charging options using the Tableau dashboard.
- Explores different brands and body styles.

Feels

- Confident when data is presented clearly through interactive visualizations.

- Satisfied after identifying the most suitable EV.

Ideation Phase
Empathize & Discover

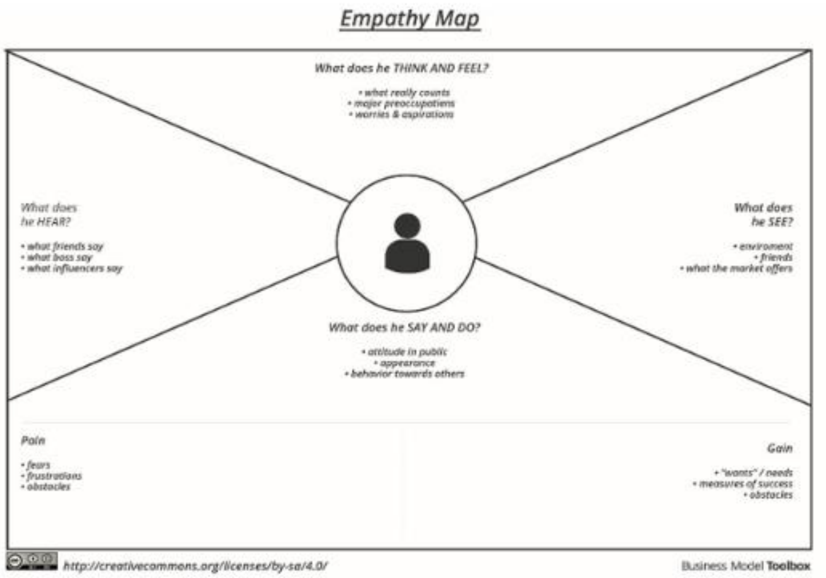
Date	12 July 2026
Team ID	
Project Name	Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

Empathy Map Canvas:

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes.

It is a useful tool to helps teams better understand their users.
Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.

Example:



2.3 Brainstorming

The brainstorming phase focused on generating innovative ideas and identifying effective approaches for analyzing Electric Vehicle (EV) data using interactive visualizations. Various challenges related to EV performance, charging infrastructure, pricing, driving range, and energy efficiency were explored to determine the most suitable analytical solution. Techniques such as **How Might We (HMW) statements, Brainstorming Ideas, Affinity Mapping, and Solution Prioritization** were used to organize ideas and select the final project approach.

Ideation Phase
Brainstorm & Idea Prioritization Template

Date	12 July 2026
Team ID	
Project Name	Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

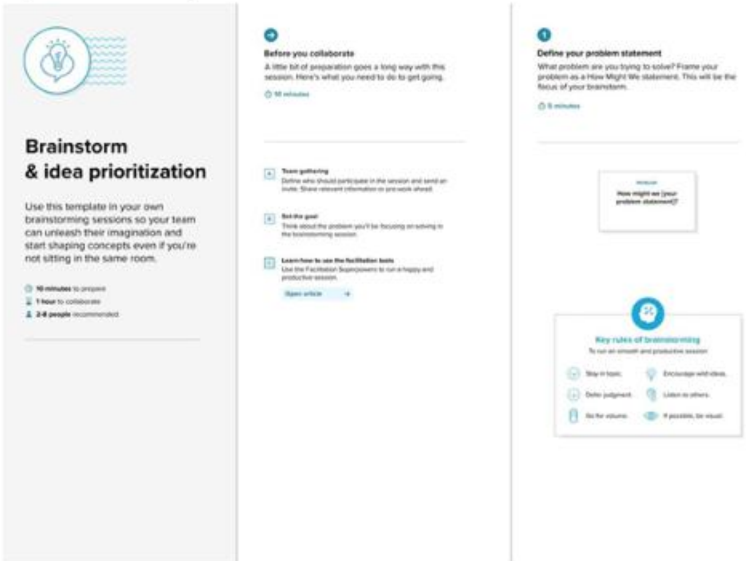
Brainstorm & Idea Prioritization Template:

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

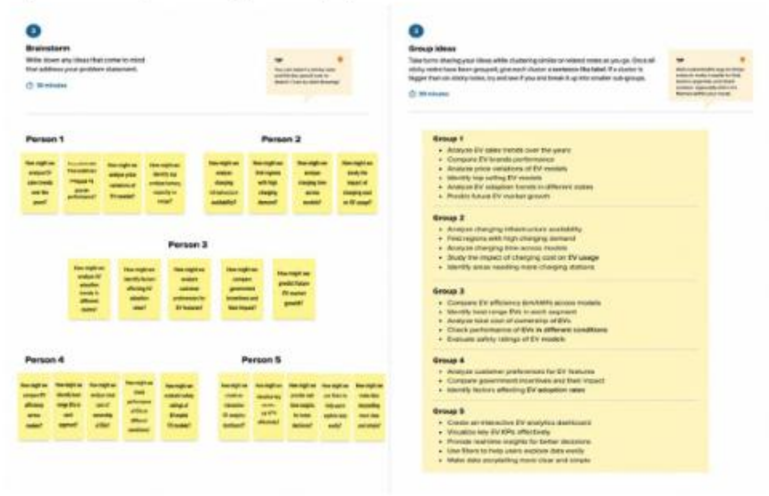
Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

Reference: https://www.mural.co/templates/brainstorm-and-idea-prioritization

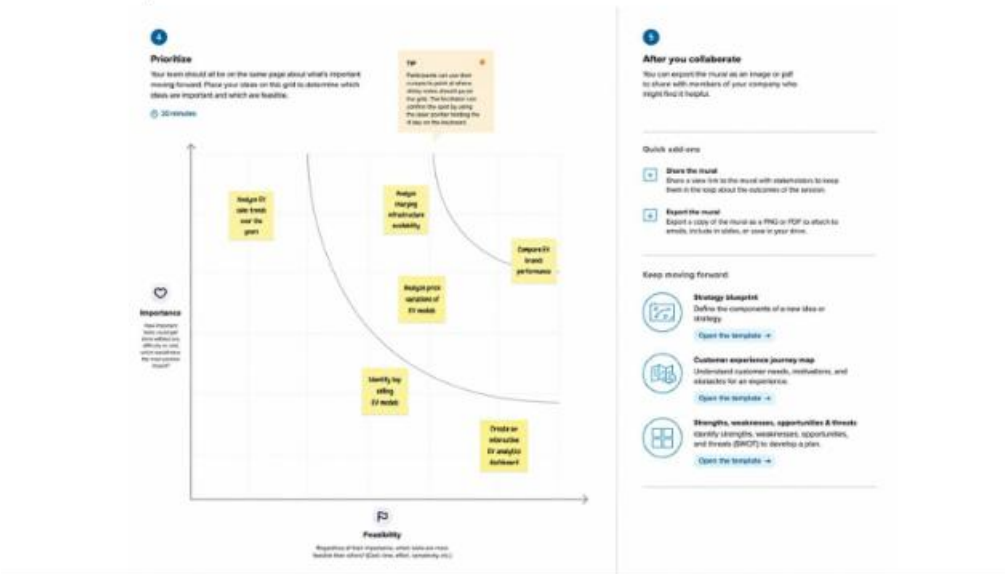
Step-1: Team Gathering, Collaboration and Select the Problem Statement



Step-2: Brainstorm, Idea Listing and Grouping



Step-3: Idea Prioritization



Conclusion

The brainstorming process helped identify the most practical and impactful solution for Electric Vehicle analysis. After evaluating multiple ideas, the team selected an interactive Tableau-based dashboard integrated with a Flask web application. This solution enables users to analyze EV models, compare prices, evaluate charging infrastructure, study vehicle efficiency, and explore market trends through interactive visualizations, supporting better understanding and data-driven decision-making.

3. REQUIREMENT ANALYSIS

The Requirement Analysis phase focused on identifying stakeholder needs, defining the functional and non-functional requirements of the Electric Vehicle Charge & Range Analysis system, and designing the overall workflow of the project. This phase ensured that the proposed solution effectively addresses the identified problem while providing an intuitive, interactive, and user-friendly platform for EV data visualization. The requirements gathered during this phase guided the development of interactive dashboards, KPI cards, filters, Tableau Stories, and Flask web integration to support efficient EV analysis and data-driven decision-making

3.1 Customer Journey Map

The Customer Journey Map illustrates the step-by-step interaction of a user while exploring the **Electric Vehicle Charge & Range Analysis** platform. It highlights the user's objectives, touchpoints, challenges, and opportunities at each stage of the journey. This analysis helped in designing an intuitive dashboard and Flask web application that simplifies EV data exploration and enhances the overall user experience.

The customer journey demonstrates how users progress from understanding the purpose of the application to exploring interactive dashboards, analyzing EV insights, comparing vehicle models, applying filters, and interpreting Tableau Story findings. Identifying user needs and challenges during this process enabled the development of a solution that provides clear visualizations, interactive analytics, and easy navigation for EV buyers, researchers, manufacturers, and policymakers.

3.2 Solution Requirement

The **Electric Vehicle Charge & Range Analysis** project was designed to transform raw electric vehicle data into meaningful visual insights through interactive dashboards and a web-based interface. The solution requirements were identified to ensure that the application provides effective data visualization, user-friendly interaction, and reliable analytical capabilities for EV buyers, researchers, manufacturers, and policymakers.

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	12 July 2026
Team ID	
Project Name	Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Home Page	Display project overview and objectives. Provide navigation to all project sections. Responsive homepage layout.
FR-2	EV Dashboard Analysis	Display interactive Tableau dashboard. Visualize EV charge, range, price, and efficiency. Support interactive filters and charts.
FR-3	EV Story Analysis	Display Tableau Story. Present EV insights sequentially. Enable navigation through story points.
FR-4	Charge & Range Analysis	Compare charging time and vehicle range. Analyze charging station availability. Generate charge and range insights.
FR-5	Brand & Price Analysis	Compare EV brands and prices. Identify top-performing brands. Display price trends and comparisons.
FR--6	Project Information	Display project overview, objectives, technologies, datasets, and developer details. Provide project documentation. Support easy navigation between sections.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The application shall provide a simple, responsive, and user-friendly interface.
NFR-2	**Security**	The application shall securely manage datasets and user interactions.
NFR-3	**Reliability**	The dashboard shall consistently display accurate visualizations and reports.
NFR-4	**Performance**	Dashboards and charts shall load quickly with minimal response time.
NFR-5	**Availability**	The application shall be accessible whenever the Tableau dashboard is published.
NFR-6	Scalability	The system shall support future datasets, dashboards, and analytical enhancements without major changes.

3.3 Data Flow Diagram

The **Data Flow Diagram (DFD)** illustrates the movement of data throughout the **Electric Vehicle Charge & Range Analysis** system. It represents how Electric Vehicle datasets are collected, preprocessed, analyzed, and transformed into interactive visualizations through Tableau. The DFD provides a clear understanding of the relationship between the EV datasets, Tableau dashboards, Flask web application, and end users.

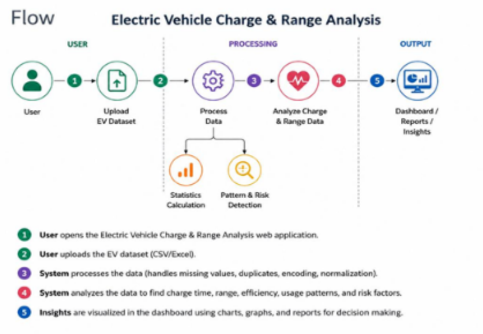
Project Design Phase-II
Data Flow Diagram & User Stories

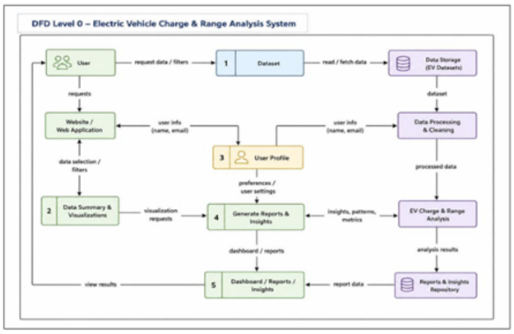
Date	12 July 2026
Team ID	
Project Name	Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: (Simplified)





User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
EV User	Dashboard	USN-1	As a user, I want to view EV dashboard insights.	Dashboard displays EV analytics	High	Sprint-1
EV User	EV Comparison	USN-2	As a user, I want to compare EV models	EV range and price comparison works	High	Sprint-1
EV User	Filters	USN-3	As a user, I want to filter EV data	Selected filters show correct results	High	Sprint-1
EV User	Charging Analysis	USN-4	As a user, I want to analyze charging stations	Charging details are displayed	Medium	Sprint-1
EV User	Range Analysis	USN-5	As a user, I want to check EV range	Range insights are generated	High	Sprint-1
Analyst	Data Analysis	USN-6	As an analyst, I want to analyze EV trends	Charts show accurate patterns	High	Sprint-2
Analyst	Reports	USN-7	As an analyst, I want to view reports	Reports display key insights	Medium	Sprint-2
Admin	Data Management	USN-8	As an admin, I want to manage EV datasets	Data can be updated easily	High	Sprint-2
EV Manufacturer	Brand Analysis	USN-9	As a manufacturer, I want to analyze EV brand performance	Brand trends and insights are displayed	Medium	Sprint-2
Researcher	Market Analysis	USN-10	As a researcher, I want to study EV market trends	EV patterns and comparisons are generated	Medium	Sprint-2
Charging Station Operator	Charging Insights	USN-11	As an operator, I want to analyze charging station usage	Charging patterns and availability are displayed	Medium	Sprint-2
EV Policy Maker	Decision Support	USN-12	As a policy maker, I want to analyze EV growth insights	EV adoption trends are identified	High	Sprint-2

3.4 Technology Stack

Tableau is used for data visualization and storytelling, providing interactive dashboards and visual analytics. The selected technologies ensure efficient data analysis, accessibility, and an enhanced user experience.

Table 3.1: Technology Stack

Project Design Phase-II
Technology Stack (Architecture & Stack)

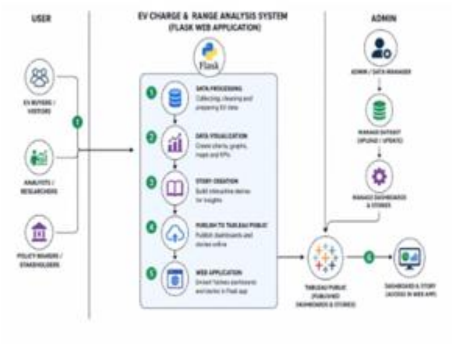
Date	12 July 2026
Team ID	
Project Name	Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	Data Source	EV dataset containing vehicle specifications, charging, range, price, and efficiency details.	CSV Dataset
2.	Data Preparation	Cleaning, formatting, and preprocessing the EV dataset before analysis.	Tableau Desktop
3.	Calculated Fields	Creation of calculated fields and KPIs for EV analysis.	Tableau Calculated Fields
4.	Data Visualization	Development of charts, graphs, maps, and analytical visualizations.	Tableau Desktop
5.	Dashboard Development	Creation of an interactive dashboard with KPIs, charts, and filters.	Tableau Dashboard
6.	Story Development	Presentation of EV insights using Tableau Story.	Tableau Story
7.	Dashboard Publishing	Publishing dashboards and stories for online access	Tableau Public
8.	User Interface	Responsive interface to access dashboards, stories, and project information.	HTML5, CSS3, Bootstrap, JavaScript
9.	Application Logic	Handles webpage navigation and embeds Tableau Dashboard and Story.	Python, Flask
10.	File Storage	Stores datasets, images, and project assets locally.	Local File System
11.	Infrastructure (Server)	Local deployment and execution of the web application	Flask Development Server

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Open-source tools used for developing the application.	Flask, Bootstrap, Tableau Public
2.	Security Implementations	Secure routing and safe embedding of Tableau dashboards.	HTML5 Validation, Flask Routing
3.	Scalable Architecture	Supports additional datasets, dashboards, and future enhancements.	Three-Tier Architecture (Data Layer, Visualization Layer, Presentation Layer)
4.	Availability	Dashboard and Story are published online and accessible anytime.	Tableau Public, Flask
5.	Performance	Optimized dashboards with responsive design and fast loading.	Tableau Public, Bootstrap, JavaScript

Technology Description

The project combines Tableau Desktop for developing interactive dashboards and Tableau Public for publishing dashboards and stories. Python is used for data handling and preprocessing, while HTML, CSS, Bootstrap, and JavaScript can be used to build a responsive interface if web deployment is required. GitHub is utilized for version control and project management, and Visual Studio Code serves as the primary development environment.

4. PROJECT DESIGN

The **Project Design** phase focuses on transforming the identified requirements into a practical and effective solution for Electric Vehicle analysis. Based on the problem statement and stakeholder needs, an interactive **Tableau-based analytics platform** was designed to simplify EV data analysis through dashboards, KPI cards, interactive visualizations, and web integration. This phase defines how the proposed solution addresses the challenges identified during the ideation and requirement analysis stages.

4.1 Problem Solution Fit

The **Electric Vehicle Charge & Range Analysis** project addresses the challenge of analyzing complex Electric Vehicle datasets by providing an interactive business intelligence solution using **Tableau**. Traditional methods of comparing EV models, charging infrastructure, pricing, and performance require manual analysis of large datasets, making it difficult to identify meaningful trends and insights. The proposed solution transforms raw EV data into interactive dashboards that enable users to analyze vehicle range, charging stations, energy efficiency, pricing, and brand performance efficiently.

The solution incorporates **one interactive dashboard, eight analytical visualizations, four KPI cards, interactive filters, and a Tableau Story** to present EV information in a structured and user-friendly manner. The solution presents EV information through interactive dashboards, KPI cards, filters, and Tableau Stories, enabling EV buyers, researchers, manufacturers, and policymakers to explore meaningful insights through an intuitive interface.

Project Design Phase
Problem – Solution Fit Template

Date	12 July 2026
Team ID	
Project Name	Electric Vehicle Charge and Range Analysis
Maximum Marks	2 Marks

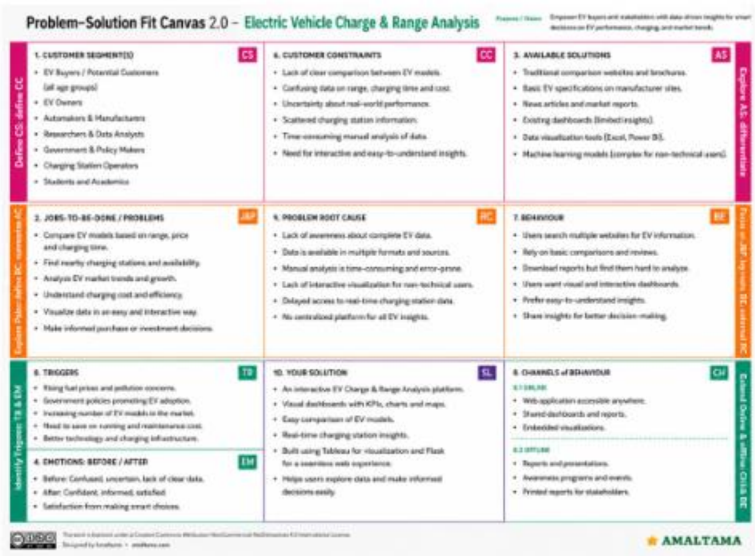
Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ❑ Solve complex problems in a way that fits the state of your customers.
- ❑ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ❑ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ❑ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ❑ **Understand the existing situation in order to improve it for your target group.**

Template:



The proposed solution successfully aligns with the identified problem by converting raw EV data into meaningful visual insights that support market analysis, vehicle comparison, charging infrastructure evaluation, and informed decision-making. The combination of interactive dashboards and web integration ensures that users can easily explore Electric Vehicle information without requiring advanced technical knowledge.

4.2 Proposed Solution

The proposed solution is an interactive **Electric Vehicle Charge & Range Analysis** platform developed using **Tableau** .The solution consolidates Electric Vehicle data, charging station information, vehicle specifications, pricing, driving range, and energy efficiency into a centralized analytical environment where users can explore EV trends through interactive dashboards and visualizations.

The platform consists of **one interactive dashboard, eight analytical visualizations, four KPI cards, interactive filters, and a Tableau Story** that collectively provide a comprehensive understanding of the Electric Vehicle ecosystem. Users can compare EV models, analyze charging infrastructure, evaluate vehicle performance, study energy efficiency, and explore pricing trends through intuitive visualizations.

Project Design Phase
Proposed Solution Template

Date	12 July 2026
Team ID	
Project Name	Electric Vehicle Charge and Range Analysis
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Electric vehicle users often find it difficult to compare EV models based on charging time, driving range, price, efficiency, and charging infrastructure. The available information is scattered across different sources, making decision-making complex. An interactive analytics platform is needed to present EV data through meaningful visualizations.
2.	Idea / Solution description	Develop an **Electric Vehicle Charge & Range Analysis** web application using **Tableau** and **Flask**. The application provides interactive dashboards, KPI cards, charts, filters, and Tableau Stories to analyze EV range, charging stations, prices, efficiency, and brand performance for better decision-making.
3.	Novelty / Uniqueness	The solution combines Tableau's interactive dashboards with a Flask web application to provide a centralized platform for EV analysis. It enables users to compare EV models, explore charging infrastructure, and understand market trends through interactive visualizations and storytelling.
4.	Social Impact / Customer Satisfaction	The application helps EV buyers, researchers, manufacturers, and policymakers make informed decisions using visual analytics. It promotes awareness of electric mobility, improves EV adoption, and simplifies the understanding of charging and range information.
5.	Business Model (Revenue Model)	The platform can be offered as a subscription-based analytics solution for automobile companies, EV manufacturers, charging station operators, research organizations, and government agencies. Additional revenue can be generated through premium dashboards, customized reports, and enterprise licensing.
6.	Scalability of the Solution	The application is designed with a scalable architecture that supports additional EV datasets, real-time charging station information

The dashboard and Tableau Story are published on **Tableau Public**, allowing users to access and interact with the visualizations through a web browser. This provides a centralized, interactive, and user-friendly platform for exploring EV analytics and supporting informed decision-making.

4.3 Solution Architecture

The **Solution Architecture** illustrates the overall structure of the **Electric Vehicle Charge & Range Analysis** system and the interaction between its major components. The architecture begins with the Electric Vehicle datasets, which serve as the primary data source. The datasets are preprocessed and analyzed in **Tableau Desktop**, where interactive dashboards, KPI cards, filters, charts, maps, and Tableau Story are developed.

The published dashboard and story are hosted on **Tableau Public**, allowing users to access them through a web browser. Users can interact with dashboards, apply filters, explore stories, and analyze insights related to EV charging infrastructure, driving range, pricing, energy efficiency, and brand performance. This architecture ensures efficient data visualization, improved accessibility, and a scalable platform for Electric Vehicle

Project Design Phase
Solution Architecture

Date	12 July 2026
Team ID	
Project Name	Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Example - Solution Architecture Diagram:

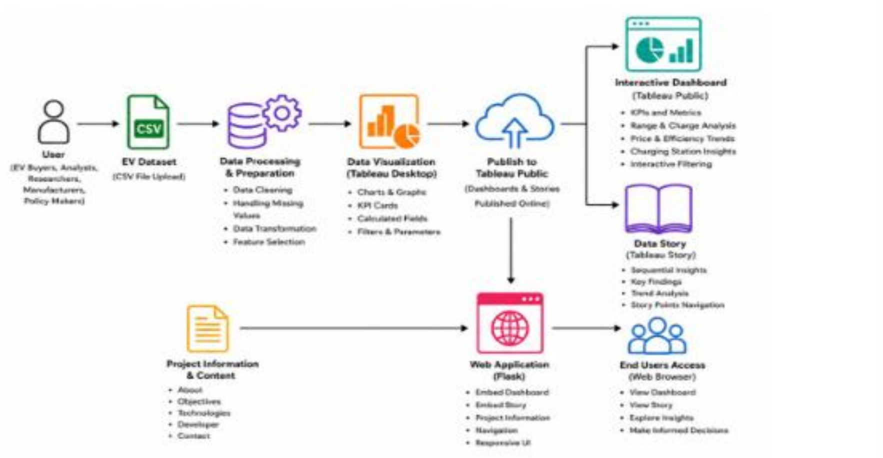


Figure 1: Architecture and data flow of the EV Charge & Range Analysis application

analysis.

The architecture demonstrates a clear flow of information from the EV datasets to the end users. By combining **Tableau** for visualization , the solution provides an interactive, user-friendly, and effective platform that supports data-driven decision-making in the Electric Vehicle domain.

5. PROJECT PLANNING & SCHEDULING

The **Project Planning and Scheduling** phase outlines the systematic execution of the **Electric Vehicle Charge & Range Analysis** project. The development process was divided into multiple sprints, enabling the team to complete dataset collection, data preprocessing, dashboard development, web integration, testing, and documentation in a structured and organized manner. The sprint-based approach ensured effective task management, continuous progress monitoring, and timely completion of project deliverables.

5.1 Project Planning

The **Electric Vehicle Charge & Range Analysis** project was planned using an iterative sprint-based approach to ensure the systematic completion of all development activities. Each sprint focused on a specific set of objectives, including dataset collection, data preprocessing, dashboard development, KPI creation, Tableau Story design, Flask web integration, testing, and documentation. This structured planning approach improved project organization, enabled continuous monitoring of progress, and ensured the timely completion of all project deliverables.

Table 5.1: Product Backlog & Sprint Planning

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	12 July 2026
Team ID	
Project Name	Electric Vehicle Charge and Range Analysis
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Preparation	USN-1	Collect and prepare the EV dataset for analysis.	3	High	Team
Sprint-1	Data Cleaning	USN-2	Clean missing values and preprocess EV data.	2	High	Team
Sprint-1	Data Visualization	USN-3	Create EV charts, graphs, and KPIs using Tableau.	5	High	Team
Sprint-2	Dashboard Development	USN-4	Develop an interactive Tableau dashboard with filters and KPI cards.	8	High	Team
Sprint-2	Story Development	USN-5	Create a Tableau Story to present EV insights sequentially.	5	High	Team
Sprint-3	Web Application	USN-6	Develop a Flask web application for the EV dashboard.	8	High	Team
Sprint-3	Dashboard Integration	USN-7	Embed the Tableau Dashboard into the Flask application.	5	High	Team
Sprint-3	Story Integration	USN-8	Embed the Tableau Story into the Flask application.	3	Medium	Team
Sprint-4	Project Documentation	USN-9	Prepare project documentation, reports, and system diagrams.	5	High	Team
Sprint-4	Testing & Deployment	USN-10	Test the application and deploy the final EV analytics system.	5	High	Team

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	15	4 Days	25 Jun 2026	28 Jun 2026	15	28 Jun 2026
Sprint-2	18	4 Days	29 Jun 2026	02 Jul 2026	18	02 Jul 2026
Sprint-3	20	4 Days	03 Jul 2026	06 Jul 2026	20	06 Jul 2026
Sprint-4	17	4 Days	07 Jul 2026	10 Jul 2026	17	10 Jul 2026

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{sprint\ duration}{velocity} = \frac{20}{10} = 2$$

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

Day	Remaining Story Points
Day 1	30
Day 2	26
Day 3	22
Day 4	18
Day 5	13
Day 6	9
Day 7	6
Day 8	3
Day 9	1
Day 10	0

Project Planning Summary

The project was executed through multiple development sprints, beginning with dataset collection and preprocessing, followed by dashboard design, visualization development, KPI creation, Tableau Story development, testing, and final documentation. The sprint-based methodology enabled efficient task allocation, continuous progress tracking, and successful completion of the Electric Vehicle Charge & Range Analysis project within the planned schedule. This structured approach ensured the timely delivery of an interactive, user-friendly, and reliable EV analytics platform that supports effective data visualization and informed decision-making.

6. FUNCTIONAL AND PERFORMANCE TESTING

The **Functional and Performance Testing** phase was conducted to verify that the **Electric Vehicle Charge & Range Analysis** project performs as expected under normal operating conditions. The testing process evaluated the functionality of the Tableau dashboard, interactive filters, KPI cards, visualizations, navigation between story points, and Tableau Story to ensure accuracy, usability, and reliability. The results confirmed that the project successfully met the intended objectives and provided meaningful analytical insights.

6.1 Functional Testing

Functional testing was performed to validate the core features of the **Electric Vehicle Charge & Range Analysis** project. Each module was tested to ensure that it functioned correctly and produced the expected results. The testing included dashboard loading, KPI card display, filter interaction, visualization accuracy, Tableau Story navigation, and accessibility of published dashboards and stories.

Table 6.1: Functional Testing

Test Case	Expected Result	Actual Result	Status
Dashboard Loading	Dashboard loads successfully	Loaded Successfully	Pass
KPI Cards Display	KPI values displayed correctly	Displayed Successfully	Pass
Dashboard Filters	Filters update visualizations correctly	Working Successfully	Pass
Charts & Maps	Visualizations display accurate data	Displayed Successfully	Pass
Tableau Story	Story opens and navigates correctly	Working Successfully	Pass
Tableau Public Dashboard	Dashboard accessible through Tableau Public	Accessible Successfully	Pass
Tableau Public Story	Story accessible through Tableau Public	Accessible Successfully	Pass

Functional Testing Summary

The functional testing results indicate that all major features of the project operate as expected. The dashboard, KPI cards, filters, visualizations, and Tableau Story function correctly without any observed issues. The published Tableau dashboard and story are accessible and provide an interactive, user-friendly experience for exploring Electric Vehicle insights.

6.2 Performance Testing

Performance testing was conducted to evaluate the responsiveness and usability of the **Electric Vehicle Charge & Range Analysis** dashboard during normal usage. The assessment focused on dashboard loading time, filter responsiveness, visualization rendering, KPI card display, and overall user interaction. The objective was to ensure that the dashboard provides accurate insights while maintaining smooth performance and a user-friendly experience.

Table 6.2: Performance Testing

Project Performance Test

Date	12 July 2026
Team ID	
Project Name	Project - Electric Vehicle Charge and Range Analysis
Maximum Marks	

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	497 Records
2.	Data Preprocessing	Data Cleaning and Data Formatting Completed
3.	Utilization of Filters	3 Main Filters Used
4.	Calculation fields Used	2 Calculated Fields used
5.	Dashboard design	8 Unique Visualizations,4 KPI Cards
6	Story Design	5 Story Points,5 Visualisations

Performance Testing Summary

The performance testing results indicate that the dashboard provides stable and reliable performance during normal usage. The dashboard loads efficiently, filters respond accurately, visualizations render correctly, and Tableau Story navigation is smooth. Overall, the project delivers an interactive and user-friendly experience for exploring Electric Vehicle data and supports effective data-driven analysis.

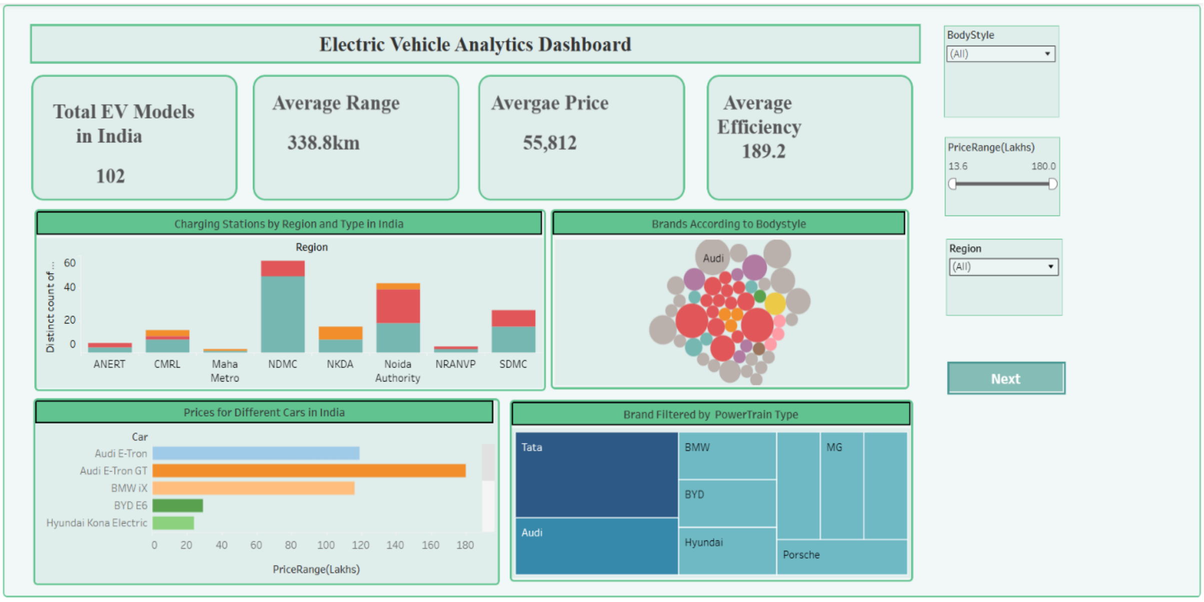
7. DOCUMENTATION & DEMO

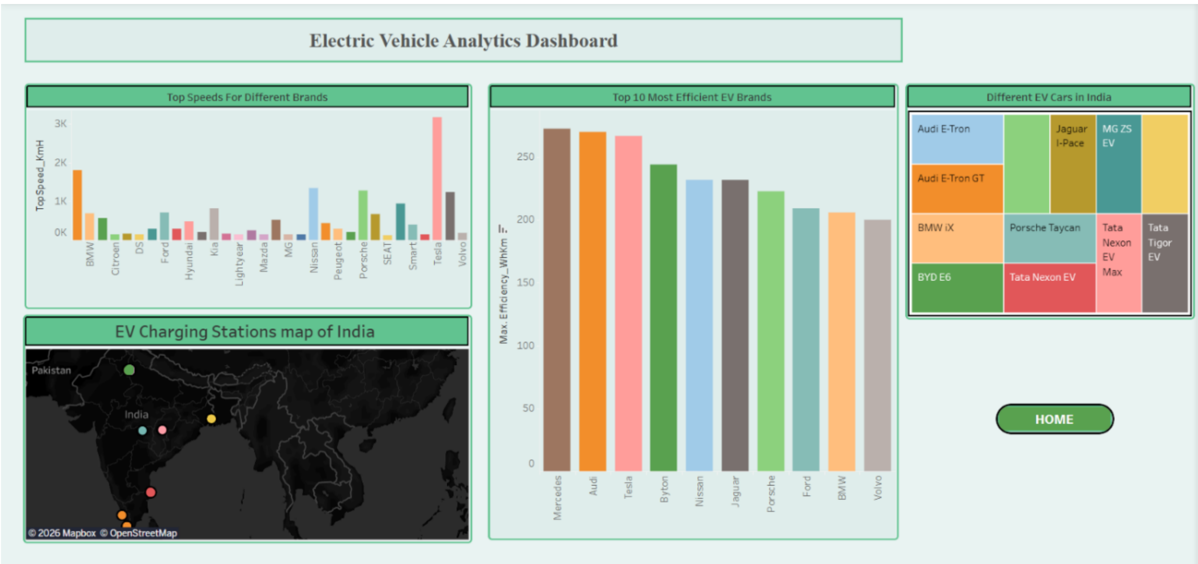
7.1 Final Dashboard

The final **Tableau Dashboard** presents a comprehensive analysis of Electric Vehicle data by combining vehicle specifications, charging infrastructure, pricing, driving range, energy efficiency, and brand information into an interactive visualization platform. The dashboard enables users to explore EV trends, compare vehicle models, analyze charging station distribution, and evaluate key performance indicators through interactive charts, maps, KPI cards, and filters.

The dashboard consists of **eight analytical visualizations** and **four KPI cards**, providing meaningful insights into the Electric Vehicle ecosystem. Users can analyze average driving range, average price, average efficiency, total EV models, charging station distribution across India, top-speed comparisons, body style distribution, and powertrain types through an intuitive and interactive interface. This dashboard simplifies complex EV data, helping users make informed and data-driven decisions.

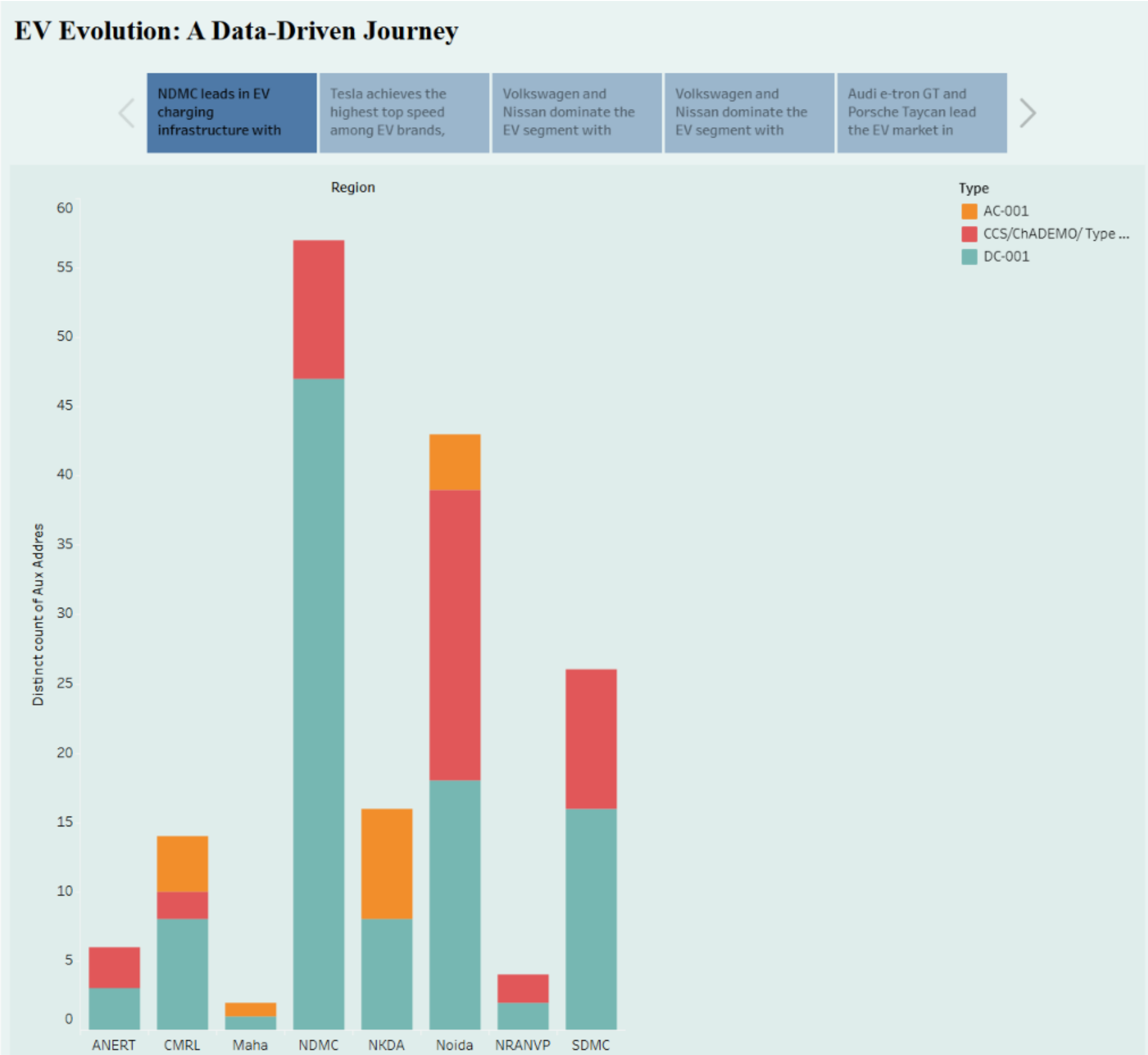
Figure 7.1: Final Tableau Dashboard





7.2 Tableau Story

The **Tableau Story** presents the analytical findings in a structured sequence, guiding users through the key insights derived from the Electric Vehicle datasets. Each story point focuses on a different aspect of EV analysis, including charging infrastructure, vehicle range, pricing, energy efficiency, brand performance, and body style distribution. This storytelling approach enables users to understand complex EV data through



7.3 GitHub Repository

The complete project source code,supporting files are maintained using GitHub for version control and collaborative development. The repository ensures organized project management, simplified maintenance, and easy access to the implementation files.

GITHUB REPOSITORY

Repository URL:

7.4 Tableau Public Links

Dashboard:

https://public.tableau.com/views/ElectricVehicleAnalysisDashboard_17839912392530/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Tableau Story:

https://public.tableau.com/views/ElectricVehicleAnalysisStory/Story1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

7.6 Project Summary

The **Electric Vehicle Charge & Range Analysis** project successfully transformed raw Electric Vehicle data into an interactive analytical platform using **Tableau**. Through **one interactive dashboard, eight visualizations, four KPI cards, interactive filters, and a Tableau Story**, the project enables users to analyze EV charging infrastructure, driving range, pricing, energy efficiency, and brand performance efficiently. The dashboard and story published on **Tableau Public** provide an accessible platform for exploring EV insights, supporting data-driven decision-making and promoting a better understanding of the Electric Vehicle ecosystem.

8. ADVANTAGES

Advantages:

- **Interactive dashboards** simplify the analysis of complex Electric Vehicle data.
- **KPI cards** provide a quick overview of important EV metrics such as range, price, efficiency, and total EV models.
- **Interactive filters** enable users to compare vehicle models, brands, body styles, and charging infrastructure dynamically.
- **Tableau Story** presents analytical findings in a structured and easy-to-understand format.
- **Geographical visualizations** help analyze the distribution of EV charging stations across different regions.
- **Published dashboards on Tableau Public** allow users to access and explore EV insights from anywhere through a web browser.

Limitations

- The analysis depends on the **quality and completeness** of the available Electric Vehicle datasets.
- The project analyzes **historical EV data** and does not provide predictive or real-time analysis.
- **Tableau Public** requires an internet connection to access the published dashboard and story.

9. CONCLUSION

The **Electric Vehicle Charge & Range Analysis using Tableau** project successfully transformed raw EV data into interactive dashboards and meaningful visual insights. Through **eight visualizations, KPI cards, interactive filters, and a Tableau Story**, the project enables users to analyze charging infrastructure, vehicle range, pricing, energy efficiency, and brand performance effectively. Overall, the project demonstrates how data visualization can support informed decision-making and improve understanding of the Electric Vehicle ecosystem.

10. FUTURE SCOPE

The current project can be enhanced further by incorporating advanced features and additional datasets. Future improvements may include:

- Integration of **real-time Electric Vehicle and charging station data**.
- Addition of **predictive analytics** for EV performance and market trends.
- Expansion of the dashboard using **larger and more diverse EV datasets**.
- Development of a **mobile-friendly web application** for improved accessibility.
- Inclusion of **AI-based recommendations** to help users choose suitable Electric Vehicles based on their preferences.

11. APPENDIX

Dataset Link

https://drive.google.com/drive/folders/1s61ViaMP8IIsGcRExO2aUpmRxDC-khEt?usp=sharing

Tableau Dashboard:

https://public.tableau.com/views/ElectricVehicleAnalysisDashboard_17839912392530/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Tableau Story:

https://public.tableau.com/views/ElectricVehicleAnalysisStory/Story1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

GitHub Repository:

https://github.com/Manasa-Builds/Electric-Vehicle-Range-Analysis

Project Demo

https://drive.google.com/file/d/1UL4JYDWUsiHwxB95uP0wEPke90G2cmUS/view?usp=sharing